

tf.compat.v1.scatter_nd_add Stay organized with collections Save and categorize content based on your preferences.

https://www.tensorflow.org/api_docs/python/tf/compat/v1/scatter_nd_add

Applies sparse addition to individual values or slices in a Variable.

Function Signatures

```
tf.compat.v1.scatter_nd_add( ref, indices, updates,  
use_locking=False, name=None )
```

Code Examples

```
tf.compat.v1.scatter_nd_add(  
ref, indices, updates, use_locking=False, name=None  
)
```

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```
[d_0, ..., d_{Q-2}, ref.shape[K], ..., ref.shape[P-1]]
```

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[d_0, ..., d_{Q-2}, ref.shape[K], ..., ref.shape[P-1]]
```

```
ref = tf.Variable([1, 2, 3, 4, 5, 6, 7, 8])
indices = tf.constant([[4], [3], [1], [7]])
updates = tf.constant([9, 10, 11, 12])
add = tf.compat.v1.scatter_nd_add(ref, indices, updates)
with tf.compat.v1.Session() as sess:
    print sess.run(add)
```

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ref = tf.Variable([1, 2, 3, 4, 5, 6, 7, 8])
indices = tf.constant([[4], [3], [1], [7]])
updates = tf.constant([9, 10, 11, 12])
add = tf.compat.v1.scatter_nd_add(ref, indices, updates)
with tf.compat.v1.Session() as sess:
    print sess.run(add)
```

```
[1, 13, 3, 14, 14, 6, 7, 20]
```

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[1, 13, 3, 14, 14, 6, 7, 20]
```

Documentation

Applies sparse addition to individual values or slices in a Variable.

ref is a Tensor with rank P and indices is a Tensor of rank Q .

indices must be integer tensor, containing indices into ref.

It must be shape $[d_0, \dots, d_{\{Q-2\}}, K]$ where $0 < K \leq P$.

The innermost dimension of indices (with length K) corresponds to indices into elements (if $K = P$) or slices (if $K < P$) along the K th dimension of ref.

updates is Tensor of rank $Q-1+P-K$ with shape:

For example, say we want to add 4 scattered elements to a rank-1 tensor to 8 elements. In Python, that addition would look like this:

The resulting update to ref would look like this:

See `tf.scatter_nd` for more details about how to make updates to slices.

Returns